Test 1

- Every solution *must* be written using complete sentences. Include all necessary reasoning in a correct order so as to construct a complete and coherent logical argument.
- You may use any definition or theorem we did in class (clearly specify what result you are using), unless the question asks you to prove something standard. Ask me in case of any doubt.
- Upload your answers on Moodle.

---

1. Let $f\colon \mathbb{R}^4 \to \mathbb{R}^3$ be defined by $f(x) = Ax$, where $A = (a_{ij})$ is an unknown 3×4 matrix. Suppose a sequence of elementary row operations transforms A into the matrix

$$R = \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

   For each of the vector spaces below, do three things. (1) State a basis for the space. (2) State a precise recipe to write a basis that is applicable to a general matrix A. (3) Provide a brief proof of your recipe. A major part of the credit is for part (3), for which a precise answer to part (2) is mandatory.

   (a) The row space of A (b) The image of f (c) The kernel of f

2. Refer to the previous problem. If you need it, let's say that the row operations used to convert A to R correspond to elementary matrices $E_1, \ldots, E_k$, in that order of application.

   (a) Find ordered bases B' of $\mathbb{R}^4$ and C' of $\mathbb{R}^3$ such that the matrix A' representing f with respect to the new bases B' and C' has the minimum possible number of off-diagonal nonzero entries. For each new basis, you must either explicitly list the vectors in it or give a precise recipe to calculate the basis vectors, possibly in terms of a matrix expressed in terms of given data.

   (b) State with proof the minimum possible number of nonzero entries in any matrix representing f.

3. Let $f\colon V \to W$ be a linear map between finite-dimensional vector spaces. Suppose f has a right inverse g_0, i.e. a linear map $g_0\colon W \to V$ such that $f \circ g_0 = \mathrm{id}_W$.

   (a) Describe in terms of g_0 the set of all right inverses of f, i.e. the set $\mathcal{R} = \{g\colon W \to V \mid f \circ g = \mathrm{id}_W\}$. A suitable subspace of one of the given vector spaces may help.

   (b) Suppose $|\mathcal{R}| > 1$ for $\mathcal{R}$ from part (a). Prove a strict inequality between $\dim V$ and $\dim W$.

4. Let V be a finite-dimensional vector space, and let $T\colon V \to V$ be a linear map.

   (a) Show that the determinant of the matrix representing T with respect to any fixed ordered basis of V is independent of the choice of basis. We can call this the determinant of T.

   (b) Let $V = M_{n\times n}$ denote the vector space of $n \times n$ matrices. Let A be a fixed $n \times n$ matrix. Define $L\colon M_{n\times n} \to M_{n\times n}$ by $L(X) = AX$. Observe (but don't prove) that L is linear. Prove that $\det(L) \neq 0$ if and only if $\det(A) \neq 0$.

   (c) Express the determinant of the linear map L in terms of A.

5. Short independent questions.

   (a) (You may do this question for credit only if you could not do Question 2 on the exam.)
   Consider $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ defined by $f(v) =$ reflection of v in the line $y = 5x$. Find a single basis B' of $\mathbb{R}^2$ such that the matrix M representing f with respect to B' has the minimum possible number of off-diagonal nonzero entries. Now find the matrix A such that $f(v) = Av$ for any $v \in \mathbb{R}^2$. You may leave your answer as a product of matrices, as long as those matrices are fully worked out.

   (b) Let $M = (m_{ij})$ be a 2026×2026 matrix. Find with justification the determinant of each of the following matrices:

$$-M, \qquad 2M, \qquad M \text{ assuming that } m_{ij} = 0 \text{ if } i < j.$$

   (c) Let $L = v_1, v_2, \ldots, v_n$ be a list of vectors in a finite-dimensional vector space V. Let $\mathbb{R}[x]$ denote the set of polynomials in one variable x with real coefficients, regarded as a vector space under the standard operations.

State and prove a necessary and sufficient condition (formulated entirely within V) on the list L such that there exists at least one linear map $f\colon V \to \mathbb{R}[x]$ satisfying $f(v_i) = x^i$ for all i.

(d) This question has two parts, both very short.
Refer to the first question on this exam. What do your recipes tell you about the dimensions of the row and column spaces of a matrix? (Optional food for thought after the exam: can one give a matrix-free proof of the answer, like we did for the rank–nullity theorem? Can one even formulate the answer in matrix-free terms? Talk to me if you are interested. Maybe we will do this later.)
Next, recall the last problem on the quiz. It asked you to show from first principles (using only the method to solve linear systems) that the rows of a $p \times q$ matrix span $\mathbb{R}^q$ if and only if the columns are independent. Now one has a very short proof of this quiz problem. How?

6. You may attempt the following problems to make up for points missed elsewhere on the exam.

(a) Let V be a vector space (not necessarily finite-dimensional). Find all linear maps $T\colon V \to V$ such that for every vector $v \in V$, the vectors v and $T(v)$ are linearly dependent.

(b) Let V be a finite-dimensional vector space. Find all linear maps $T\colon V \to V$ that commute with every linear map $S\colon V \to V$, meaning that

$$T \circ S = S \circ T \qquad \text{for all linear maps } S\colon V \to V.$$

(Extra, optional: is your answer valid if V is infinite-dimensional?)